

Correctness of the boolean **Decompose** circuit

ML-DSA, $\gamma_2 = (q - 1)/32$: 43 AND gates per coefficient

1 Setting

Throughout, $q = 8380417$ and

$$\gamma_2 = \frac{q-1}{32} = 261888, \quad \alpha = 2\gamma_2 = 523776 = 2^9 \cdot 1023, \quad 16\alpha = q - 1.$$

The input is an integer $r \in [0, 2^{23})$ held as 23 boolean-shared bits $r[0..22]$ (little endian). Step 4 of Π_{PrepSign} must output $w_1 = \text{HighBits}(r)$ in the clear and keep $w_0 = \text{LowBits}(r)$ shared, where $(w_1, w_0) = \text{Decompose}_q(r, \alpha)$ is the FIPS 204 routine.

The circuit is stated in terms of the following slices of r , all of which are free (pure rewiring):

$$L = r \bmod 2^9 = r[0..8], \quad R_{\text{lo}} = r[9..18], \quad R_{\text{hi}} = \lfloor r/2^{19} \rfloor = r[19..22] \in [0, 16).$$

We write $\rho = \lfloor r/2^9 \rfloor$, so that $r = 2^9 \rho + L$ and $\rho = 1024 R_{\text{hi}} + R_{\text{lo}}$.

2 Removing the FIPS branch

FIPS 204 defines **Decompose** by cases. We first replace it by a single floor function plus one correction bit.

Lemma 1 (centred quotient). *Let r_1^* be the unique integer with $r_0^* := r - \alpha r_1^* \in (-\alpha/2, \alpha/2]$. Then*

$$r_1^* = \left\lfloor \frac{r + \alpha/2 - 1}{\alpha} \right\rfloor.$$

Proof. $-\alpha/2 < r - \alpha r_1 \leq \alpha/2$ is equivalent to $\frac{r - \alpha/2}{\alpha} \leq r_1 < \frac{r + \alpha/2}{\alpha}$, whose unique integer solution is $r_1^* = \lceil (r - \alpha/2)/\alpha \rceil = \lfloor (r + \alpha/2 - 1)/\alpha \rfloor$. $\square$

The offset is $\alpha/2 - 1$, not $\alpha/2$: the interval is half-open on the left, so $r = \alpha/2$ must give $r_1^* = 0$ rather than 1.

Lemma 2 (the branch is a mask and a decrement). *For $r \in [0, q)$ one has $r_1^* \in [0, 16]$, and the FIPS 204 output (r_1, r_0) satisfies*

$$r_1 = r_1^* \bmod 16, \quad r_0 = r_0^* - \text{sp}, \quad \text{sp} := \lfloor r_1^* = 16 \rfloor.$$

Proof. FIPS 204 returns (r_1^*, r_0^*) unless $r - r_0^* = q - 1$, in which case it returns $(0, r_0^* - 1)$. Since $r - r_0^* = \alpha r_1^*$ and $16\alpha = q - 1$, that condition is exactly $r_1^* = 16$; and $16 \bmod 16 = 0$. $\square$

Lemma 2 is what makes the circuit cheap: the data-dependent branch has become a truncation (free) plus a single bit **sp**.

3 Dividing by $\alpha = 2^9 \cdot 1023$

Lemma 3 (splitting off 2^9). *Let $g = \lfloor L \geq 257 \rfloor$. Then $r_1^* = \left\lfloor \frac{\rho + 511 + g}{1023} \right\rfloor$.*

Proof. Since $\alpha/2 - 1 = 261887 = 511 \cdot 512 + 255$,

$$r + \alpha/2 - 1 = 512(\rho + 511) + (L + 255), \quad \text{hence} \quad \frac{r + \alpha/2 - 1}{\alpha} = \frac{\rho + 511 + \frac{L+255}{512}}{1023}.$$

Write $\frac{L+255}{512} = g + f$ with $g = \lfloor \frac{L+255}{512} \rfloor = \lfloor L \geq 257 \rfloor$ and $f \in [0, 1)$. For an integer A , an integer $M \geq 1$ and $f \in [0, 1)$ we have $\lfloor (A + f)/M \rfloor = \lfloor A/M \rfloor$: if $kM \leq A + f < A + 1$ then $kM \leq A$, so the two floors bound each other. Apply this with $A = \rho + 511 + g$ and $M = 1023$. $\square$

The rounding offset never becomes an addition: $\alpha/2 - 1$ splits into 511 (absorbed into the dividend) and 255 (which turns into the threshold $L \geq 257$), and a threshold is a choice of wire.

Lemma 4 (casting out nines; HighBits). *Let $n := R_{\text{lo}} + R_{\text{hi}} + g$ and $h := \lfloor n \geq 512 \rfloor$. Then $r_1^* = R_{\text{hi}} + h$.*

Proof. Because $1024 = 1023 + 1$, $\rho = 1024R_{\text{hi}} + R_{\text{lo}} = 1023R_{\text{hi}} + (R_{\text{hi}} + R_{\text{lo}})$, so Lemma 3 gives

$$r_1^* = R_{\text{hi}} + \left\lfloor \frac{R_{\text{hi}} + R_{\text{lo}} + g + 511}{1023} \right\rfloor = R_{\text{hi}} + \left\lfloor \frac{n + 511}{1023} \right\rfloor.$$

Now $0 \leq n \leq 1023 + 15 + 1 = 1039$, hence $511 \leq n + 511 \leq 1550 < 2046 = 2 \cdot 1023$: the quotient is 0 or 1, and it is 1 iff $n + 511 \geq 1023$, i.e. iff $n \geq 512$. $\square$

Remark 1. $n \leq 1039 < 1536$, so $n_{10} = 1$ forces $n \in [1024, 1039]$ and $n_9 = 0$. The two bits are disjoint and $h = n_9 \vee n_{10} = n_9 \oplus n_{10}$ is free under free-XOR.

4 LowBits from the same intermediates

The naive route computes $r_0^* = r - \alpha r_1^*$ with a constant multiply and a 23-bit subtraction. It is unnecessary: r_0^* is already present in the quantities Lemma 4 produced.

Theorem 1. $r_0^* = 512(n - g - 1023h) + L$.

Proof. Using $R_{\text{hi}} + R_{\text{lo}} = n - g$ and $\rho = 1023R_{\text{hi}} + (R_{\text{hi}} + R_{\text{lo}})$,

$$\begin{aligned} r_0^* &= r - \alpha r_1^* = 512\rho + L - 512 \cdot 1023(R_{\text{hi}} + h) \\ &= 512 \left[(1023R_{\text{hi}} + (R_{\text{hi}} + R_{\text{lo}})) - 1023R_{\text{hi}} - 1023h \right] + L \\ &= 512 \left[(n - g) - 1023h \right] + L. \quad \square \end{aligned}$$

Write $m := n - g - 1023h$, so that $r_0^* = 512m + L$.

Lemma 5 (range). $m \in [-512, 511]$, which is exactly the range of a 10-bit two's complement word.

Proof. If $h = 0$ then $n \leq 511$ and $m = n - g = R_{\text{hi}} + R_{\text{lo}} \geq 0$, so $m \in [0, 511]$. If $h = 1$ then $512 \leq n \leq 1039$ and $m = n - g - 1023 \in [512 - 1 - 1023, 1039 - 0 - 1023] = [-512, 16]$. $\square$

Consequently $r_0^* = 512m + L$ with $L \in [0, 512)$ is precisely the 19-bit two's complement word $m \parallel L$, by the standard fact that a k -bit two's complement high word concatenated with a j -bit unsigned low word denotes $2^j m + L$ in $(k + j)$ -bit two's complement. (Consistency check: $r_0^* \in (-\gamma_2, \gamma_2] \subset [-2^{18}, 2^{18})$.)

Corollary 1 (the 1023 disappears). *Only $m \bmod 2^{10}$ is needed, and $-1023 \equiv 1 \pmod{1024}$, so*

$$m \equiv n - g + h \pmod{1024}.$$

The multiplication by 1023 therefore never happens, and only the low ten bits of n are used.

5 Paying for the branch

By Lemma 2, $r_1 = (R_{\text{hi}} + h) \bmod 16$: the 4-bit incrementer simply does not compute the carry out of bit 3, which is why **HighBits** alone gets the special case for free. That discarded carry is

$$\text{sp} = [R_{\text{hi}} + h = 16] = R_{\text{hi}}[0] \wedge R_{\text{hi}}[1] \wedge R_{\text{hi}}[2] \wedge R_{\text{hi}}[3] \wedge h,$$

the last link of the same carry chain: keeping it costs one AND gate.

Lemma 6 (re-normalising the -1). *Let $b := [L < \text{sp}] = \text{sp} \wedge [L = 0]$ and $L' := (L - \text{sp}) \bmod 512$. Then*

$$r_0 = r_0^* - \text{sp} = 512(m - b) + L', \quad m - b \in [-512, 511].$$

Proof. The identity is the definition of b and L' , since $L - \text{sp} = -512b + L'$. For the range it suffices to rule out $m = -512$ when $b = 1$. Now $b = 1$ forces $\text{sp} = 1$, hence $h = 1$ and $R_{\text{hi}} = 15$; it also forces $L = 0$, hence $g = [0 \geq 257] = 0$. Therefore $n = R_{\text{lo}} + 15$, and $n \geq 512$ gives $R_{\text{lo}} \geq 497$, so $m = R_{\text{lo}} + 15 - 1023 \geq -511$. $\square$

In the circuit b and L' come from the borrow chain $b_0 = \text{sp}$, $b_{i+1} = b_i \wedge \neg L_i$, $L'_i = L_i \oplus b_i$, whose output $b_9 = \text{sp} \wedge \neg L_0 \wedge \dots \wedge \neg L_8 = \text{sp} \wedge [L = 0]$ is the claimed b .

Lemma 7 (the three ± 1 corrections fuse). *By Corollary 1 and Lemma 6, $m - b \equiv n + s \pmod{1024}$ with $s := h - g - b \in [-2, 1]$. The 2-bit two's complement encoding (s_1, s_0) of s , i.e. $s = -2s_1 + s_0$, is*

$$s_0 = h \oplus g \oplus b, \quad s_1 = [s < 0] = \begin{cases} g \wedge b, & h = 1, \\ g \vee b, & h = 0, \end{cases} \quad \text{i.e.} \quad s_1 = (g \vee b) \oplus (h \wedge ((g \vee b) \oplus (g \wedge b))).$$

Proof. Exhaustively, $s = 1, 0, -1, -2$ encode as $(0, 1), (0, 0), (1, 1), (1, 0)$ and $-2s_1 + s_0$ returns $1, 0, -1, -2$. The bit s_0 is the parity of $h + g + b$. For s_1 : if $h = 0$ then $s = -(g + b) < 0$ iff $g \vee b$; if $h = 1$ then $s = 1 - g - b < 0$ iff $g \wedge b$. $\square$

Sign-extending (s_1, s_0) to ten bits (repeat s_1 in bits 1..9) and adding it to n with a ripple chain whose final carry is dropped realises $m - b \bmod 2^{10}$, which by Lemma 5 is $m - b$ itself.

Gate count. g : 8; n : 10; h : 0; w_1 and sp : 4; the borrow chain for L' : 9; s : 3; the 10-bit add: 9. Total 43 AND gates, against 21 for **HighBits** alone. Exhaustively verified against FIPS 204 for every $r \in [0, q)$.

6 Why $\mathbf{sp}$ must not be revealed

Revealing $\mathbf{sp}$ would save the ten gates of Lemma 6 and Lemma 7, because $w_0 = w - \alpha w_1 - \mathbf{sp}$ is then an affine function of w with public coefficients and can be computed locally on arithmetic shares. It is exactly the one bit that may not be leaked.

Lemma 8. *Conditioned on the public value $w_1 = 0$, we have $\mathbf{sp} = [w_0 < 0]$.*

Proof. By Lemma 2, $w_1 = 0$ holds iff $r_1^* \in \{0, 16\}$. If $r_1^* = 0$ then $w_0 = r \in [0, \gamma_2]$ and $\mathbf{sp} = 0$. If $r_1^* = 16$ then $w_0 = r - (q - 1) - 1 \in [-\gamma_2, -1]$ and $\mathbf{sp} = 1$. $\square$

$\mathcal{F}_{\text{PrepSign}}$ outputs w_1 in the clear and w_0 only in shared form. Publishing $\mathbf{sp}$ would additionally output $\text{sign}(w_0)$, a nontrivial predicate of $w = \mathbf{A}\mathbf{y} + \mathbf{e}_w$ and hence of the secret nonce $\mathbf{y}$; no simulator given w_1 alone can produce it.

7 ML-DSA-44

Here $\alpha = 190464 = 2^{11} \cdot 93$, $\gamma_2 = 95232$, $44\alpha = q - 1$ and $\alpha/2 - 1 = 95231 = 46 \cdot 2048 + 1023$. With $L = r \bmod 2^{11}$ and $\rho = \lfloor r/2^{11} \rfloor$, the arguments of Lemmas 3 and 4 give

$$g = \lfloor L \geq 1025 \rfloor, \quad t = \rho + 46 + g, \quad r_1^* = \lfloor t/93 \rfloor.$$

93 is not a Mersenne number, but $11 \cdot 93 = 1023$, so identically $\lfloor t/93 \rfloor = \lfloor 11t/1023 \rfloor$. Writing $u = 11t = 1024a + b$ with $b = u \bmod 1024$, the same fold $u = 1023a + (a + b)$ together with $a + b \leq 44 + 1023 < 2 \cdot 1023$ yields

$$r_1^* = a + h, \quad h = \lfloor a + b \geq 1023 \rfloor.$$

The gift of Theorem 1 does not survive the pre-multiplication by 11: one finds $\rho - 93r_1^* = \frac{a+b-1023h-506}{11} - g$, and the exact division by 11 is more expensive than recomputing. So the circuit evaluates r_0 directly,

$$r_0 = r - 2^{11} \cdot (93r_1^*) - f, \quad f = \lfloor r_1^* = 44 \rfloor,$$

using $93x = 31 \cdot (3x) = 32(3x) - (3x)$. Since $|r_0| \leq \gamma_2 < 2^{17}$, the result is an 18-bit two's complement word — a 7-bit high word above the 11 bits of L' — so $93r_1^*$ is only ever needed modulo 2^7 ; truncation is legitimate because reduction mod 2^7 is a ring homomorphism and the top carry is discarded anyway. Finally $-D - b = \neg D + \neg b$ for a bit b (as $\neg D = 2^7 - 1 - D$ and $1 - b = \neg b$), so the borrow enters as the carry-in of a single ripple add. The cost is $71 + 29 = 100$ AND gates, again exhaustively verified over $r \in [0, q)$.